Linguistic Relativity as an Existence Result

A Constructive Proof that Coding Schemes Induce Non-Isometric Geometries

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Abstract

A *language* is a coding scheme: a stochastic mapping from semantic states to strings. Different coding schemes may encode the same meanings differently—compositionally, by assembling expressions from atomic units, or lexically, by assigning dedicated symbols to frequent combinations.

A *predictive learner* trained on a corpus of strings develops internal representations optimized for predicting continuations. These representations inherit a geometry: internal states are *close* if they induce similar output distributions, *distant* if they predict differently. We formalize distance via Jensen–Shannon divergence between the distributions over next tokens.

We construct two coding schemes L_A and L_B over a shared anchored domain—a set of meanings with identity criteria external to any encoding. Both languages cover the domain: every anchored meaning is expressible in both. Corpora are generated from identical distributions over meanings. Yet we prove: any predictive learner that achieves adequate prediction accuracy on these corpora must induce *non-isometric* representational geometries. There exist pairs of anchored meanings whose representational distance differs between the two induced geometries, despite the meanings themselves being identical across languages.

The result is constructive. We exhibit the coding schemes, derive the geometric divergence from their structure, and show that non-isometry is a necessary consequence of how each language factors meaning into expressible units—not an artifact of learner architecture or training dynamics. The implication: representational geometry is shaped by the structure of the code, not merely by what is encoded.

1. The Claim

The question is: *when does the structure of a coding scheme determine the geometry of learned representations?*

A predictive learner trained on text develops internal states—representations—that support prediction. These representations have geometric structure: some pairs of states are similar (they induce similar output distributions), others are distant (they predict differently). This geometry is not arbitrary; it reflects the statistical structure the learner has extracted from its training corpus.

Now consider two different coding schemes over the same anchored domain—a domain whose meanings have identity criteria independent of how any language encodes them. Both can express every meaning in the domain; corpora are drawn from identical distributions over meanings. The only difference is *how* meanings are encoded into strings. Must learners trained on these corpora develop the same representational geometry?

The answer is no. We prove:

Theorem 1 (Non-Isometry from Coding Structure)**.** *There exist coding schemes L_A and L_B over an anchored domain $\mathcal{A}$ satisfying:*

(i) ***Coverage****: Every meaning $a \in \mathcal{A}$ is expressible in both L_A and L_B.*

(ii) ***Distributional identity****: Corpora are generated by sampling meanings $a \sim P_{\mathcal{A}}$ and encoding them as strings $x \sim P_L(x \mid a)$.*

(iii) ***Behavioral geometry****: A predictive learner T achieving cross-entropy loss below ε on corpus $\mathcal{C}_L$ induces an internal geometry $(\mathcal{H}_L, d_L)$ where $d_L(h_1, h_2) = \mathrm{JS}\big(p(\cdot \mid h_1), p(\cdot \mid h_2)\big)$.*

Such that the induced geometries are non-isometric: there exist anchored meanings $a_1, a_2 \in \mathcal{A}$ with

$$d_{L_A}\big(h(a_1), h(a_2)\big) \neq d_{L_B}\big(h(a_1), h(a_2)\big)$$

where $h(a)$ denotes the learner's internal state after processing a canonical expression for a.

Conditions (i)–(iii) define the setup; the non-isometry is the conclusion. The theorem is constructive: we exhibit L_A and L_B, identify the witness pair (a_1, a_2), and derive the inequality from the coding structure alone.

We prove Theorem 1 via two constructions. The first is minimal: one language collapses two meanings into a single expression while the other keeps them distinct, yielding $d > 0$ versus $d = 0$. This suffices for existence but invites a triviality objection (same string implies same state by definition). The second construction is non-degenerate: compositional versus lexicalized encodings that both distinguish all meanings, yet differ in *hub structure*—shared intermediate states that lie on processing paths to multiple meanings. The stronger result (Theorem 30, Section 6) shows the phenomenon persists even without collapse, as a difference in processing dynamics rather than mere identity.

1.1 Anchored Domains

The theorem requires that we can identify "the same meaning" across two different coding schemes. This is non-trivial. If concepts are defined by their position in a representational manifold, and the manifolds differ, how do we establish correspondence?

We avoid circularity by restricting to *anchored domains*: sets of meanings with identity criteria external to any particular encoding.

Definition 2 (Anchored Domain)**.** *An* anchored domain $(\mathcal{A}, \sim)$ *consists of a set $\mathcal{A}$ of meanings together with an equivalence relation $\sim$ defined by criteria independent of any coding scheme. Two meanings $a_1, a_2 \in \mathcal{A}$ are identified as "the same" or "different" by the anchoring structure, not by how languages encode them.*

For kinship, the anchoring is genealogical: positions are defined by biological and social relations (parent-of, sibling-of, spouse-of). "Mother's brother" is a position in a kinship graph—the same position whether we call it *uncle*, *māmā*, or encode it compositionally as MOTHER + SIBLING + MALE. The anchoring is the graph structure, not the linguistic expression.

Definition 3 (Coverage). *A coding scheme L* covers *an anchored domain $\mathcal{A}$ if for every $a \in \mathcal{A}$, there exists at least one expression $x \in \Sigma^*$ such that x denotes a in L.*

Coverage says: the language can talk about everything in the domain. It does not say the language can express "all possible concepts"—that would require a theory of what concepts exist, which we avoid. We construct a specific anchored domain and require coverage of that domain.

The theorem's force: two languages covering the same anchored domain, generating corpora from identical distributions over that domain, can nonetheless induce non-isometric geometries. The difference arises from *how* meanings are encoded—compositionally versus lexically—not from which meanings are available.

1.2 The Argument Structure

The proof proceeds as follows:

1. **Setup (Section 2).** Define coding schemes, corpora, predictive learners, and behavioral geometry.
2. **Minimal Construction (Section 3).** Build a kinship domain and two languages: L_D (distinguished encoding) and L_C (collapsed encoding, where multiple meanings share an expression).
3. **Key Lemma (Section 4).** Prove that an ε-competent learner must assign different output distributions to contextually-differentiated tokens.
4. **Minimal Non-Isometry (Section 5).** Apply the lemma: $d > 0$ for distinguished meanings, $d = 0$ for collapsed meanings. This proves Theorem 1.
5. **Strong Construction (Section 6).** Build compositional (L_{compl}) versus lexicalized (L_{lex}) encodings. Both distinguish all meanings; neither collapses. Prove Theorem 30: the geometries differ in hub structure and processing dynamics.
6. **Scope (Section 7).** Clarify what the results do and do not establish.

The constructions are deliberately minimal. We use a toy semantic domain with a handful of meanings. The proof requires only that the learner achieve reasonable prediction accuracy—no claims about architecture, capacity, or training dynamics beyond competent language modeling.

2. Setup

We now define the formal objects: coding schemes, corpora, predictive learners, and behavioral geometry.

2.1 Coding Schemes

Definition 4 (Coding Scheme). *A* coding scheme *L over an anchored domain $\mathcal{A}$ consists of:*

(a) A finite alphabet Σ

(b) For each meaning $a \in \mathcal{A}$, a non-empty set of valid expressions $E_L(a) \subseteq \Sigma^*$

(c) A conditional distribution $P_L(x \mid a)$ over expressions, supported on $E_L(a)$

The distribution $P_L(x \mid a)$ captures that speakers may express the same meaning in multiple ways with different probabilities. For a compositional language, the high-probability expressions might be systematic combinations of morphemes. For a lexicalized language, a single fused token might dominate.

We do not require that expressions be unambiguous. A string x may express multiple meanings: $x \in E_L(a_1) \cap E_L(a_2)$ is permitted. Context typically resolves ambiguity in natural language; our construction will avoid it for clarity.

2.2 Corpora

Definition 5 (Corpus Generation). *Given an anchored domain $\mathcal{A}$ with distribution $P_{\mathcal{A}}$ over meanings, and a coding scheme L, a* corpus *$\mathcal{C}_L$ is generated by:*

(a) Sample meaning: $a \sim P_{\mathcal{A}}$

(b) Sample expression: $x \sim P_L(x \mid a)$

(c) Emit x (the meaning a is not observed)

Repeat to generate a sequence of strings.

The corpus is a sequence of strings without meaning labels. The learner sees only the surface forms, not the underlying meanings that generated them. This models the situation of a language learner exposed to text: the semantic intentions of speakers are latent.

For the theorem, both L_A and L_B generate corpora from the *same* distribution $P_{\mathcal{A}}$ over meanings. The languages differ only in how meanings map to strings—the encoding $P_L(x \mid a)$—not in which meanings are discussed.

2.3 Predictive Learners

Definition 6 (Predictive Learner). *A* predictive learner *T is a system that, given a corpus $\mathcal{C}$ of strings, learns to model the distribution over continuations. Formally, T induces:*

(a) A set of internal states $\mathcal{H}$

(b) A mapping from string prefixes to internal states: $h : \Sigma^* \to \mathcal{H}$

(c) A distribution over next tokens given internal state: $p_T(\sigma \mid h)$ for $\sigma \in \Sigma$

We impose no constraints on the architecture of T—it could be a transformer, an RNN, an n-gram model, or any other system. The only requirement is *predictive competence*:

Definition 7 (Predictive Competence). *A learner T is* ε-competent *on corpus $\mathcal{C}_L$ if its cross-entropy loss is bounded:*

$$H(P_L, p_T) = -\mathbb{E}_{x \sim P_L}\big[\log p_T(x)\big] \le H(P_L) + \varepsilon$$

where $H(P_L)$ is the entropy of the true distribution and $p_T(x) = \prod_i p_T(x_i \mid x_{<i})$.

An ε-competent learner models the corpus distribution to within ε nats of optimal. This is a weak requirement: it says only that the learner has extracted the statistical regularities of the corpus well enough to predict competently. It does not require that the learner "understand" meaning, represent concepts explicitly, or have any particular internal structure.

The theorem will show that *any* ε-competent learner—regardless of architecture—must develop certain geometric properties. The non-isometry is forced by the coding structure, not by assumptions about how the learner works.

2.4 Behavioral Geometry

The internal states $\mathcal{H}$ of a learner have no intrinsic coordinate system. Different learners may use entirely different representations. To compare geometries across learners and across languages, we need a *coordinate-free* metric grounded in observable behavior.

Definition 8 (Behavioral Distance)**.** *The* behavioral distance *between internal states* $h_1, h_2 \in \mathcal{H}$ *is:*

$$d(h_1, h_2) = \mathrm{JS}\big(p_T(\cdot \mid h_1),\, p_T(\cdot \mid h_2)\big)$$

where JS *is the Jensen–Shannon divergence:*

$$\mathrm{JS}(p, q) = \frac{1}{2} D_{\mathrm{KL}}(p \| m) + \frac{1}{2} D_{\mathrm{KL}}(q \| m), \quad m = \frac{p+q}{2}$$

Jensen–Shannon divergence is symmetric, bounded ($0 \le \mathrm{JS} \le \ln 2$), and zero if and only if the distributions are identical. Its square root is a metric.

Interpretation. Two internal states are *close* if they induce similar predictions about what comes next. They are *distant* if they predict differently. This grounds distance in behavior: we don't ask what the internal representation "looks like," only how it functions as a predictor.

This is the key move that makes cross-linguistic comparison possible. We cannot compare the raw internal states of two different learners—they may live in entirely different spaces. But we can compare their behavioral geometries: the pattern of distances defined by predictive similarity.

2.5 From Meanings to Internal States

The theorem concerns distances between representations of *meanings*. But the learner never sees meanings directly; it sees strings. We need a mapping from meanings to internal states.

Definition 9 (Canonical Expression)**.** *For meaning* $a \in \mathcal{A}$ *in language* L*, the* canonical expression $x^*_L(a)$ *is the maximum-probability expression:*

$$x^*_L(a) = \arg\max_{x \in E_L(a)} P_L(x \mid a)$$

Definition 10 (Meaning Representation)**.** *The* representation of meaning *a in a learner* T *trained on* L *is:*

$$h_L(a) = h\big(x^*_L(a)\big)$$

the internal state reached after processing the canonical expression for a*.*

This definition ties meaning to representation via the most typical linguistic expression. Alternative definitions (averaging over all expressions, using context-embedded expressions) are possible; we use canonical expressions for simplicity. The key results are robust to this choice.

2.6 The Geometric Claim, Precisely

We can now state precisely what the theorem claims.

Let $\mathcal{A}$ be an anchored domain. Let L_A and L_B be coding schemes over $\mathcal{A}$, both covering $\mathcal{A}$, generating corpora from the same distribution $P_{\mathcal{A}}$. Let T_A and T_B be ε-competent learners on $\mathcal{C}_{L_A}$ and $\mathcal{C}_{L_B}$ respectively.

Each learner induces a geometry on $\mathcal{A}$:

$$d_{L_A}(a_1, a_2) = d\big(h_{L_A}(a_1), h_{L_A}(a_2)\big) \tag{1}$$
$$d_{L_B}(a_1, a_2) = d\big(h_{L_B}(a_1), h_{L_B}(a_2)\big) \tag{2}$$

The theorem asserts: there exist L_A, L_B, and anchored meanings a_1, a_2 such that for *any* ε-competent learners:

$$d_{L_A}(a_1, a_2) \neq d_{L_B}(a_1, a_2)$$

The geometries are non-isometric. No distance-preserving map can align them. The same meanings, at different representational distances, because the codes differ.

3. The Construction

We now construct the anchored domain and two coding schemes that satisfy the theorem's conditions.

3.1 The Kinship Domain

Kinship provides an ideal anchored domain. Kinship positions are defined by genealogical relations—biological facts (who gave birth to whom) and social facts (who married whom)—that exist independently of how any language encodes them.

Definition 11 (Kinship Graph)**.** *A* kinship graph $G = (V, E)$ *is a directed labeled graph where:*

- *Vertices V represent individuals*
- *Edges E are labeled by relation type:* PARENT-OF, SIBLING-OF, SPOUSE-OF

Definition 12 (Kinship Position)**.** *Given a kinship graph G and a designated ego vertex e, a* kinship position *is a path type from ego to another vertex. Positions are defined by the sequence of relation types traversed.*

For example:

- MOTHER: ego $\xleftarrow{\text{parent-of}}$ female
- FATHER: ego $\xleftarrow{\text{parent-of}}$ male
- MOTHER'S-BROTHER: ego $\xleftarrow{\text{parent-of}}$ female $\xleftarrow{\text{sibling-of}}$ male
- FATHER'S-BROTHER: ego $\xleftarrow{\text{parent-of}}$ male $\xleftarrow{\text{sibling-of}}$ male

The anchoring is the graph structure. "Mother's brother" refers to the same genealogical position regardless of whether a language calls it *uncle*, *māmā*, or encodes it compositionally. Two speakers with the same family tree, speaking different languages, refer to the same person when they use their respective terms for this position.

Definition 13 (The Anchored Kinship Domain)**.** *Let $\mathcal{A}_{\text{kin}}$ be the set of kinship positions reachable within two steps from ego:*

$$\mathcal{A}_{\text{kin}} = \{\textsc{Mo}, \textsc{Fa}, \textsc{MoBr}, \textsc{FaBr}, \textsc{MoSi}, \textsc{FaSi}, \textsc{Br}, \textsc{Si}\}$$

with identity given by genealogical path type.

This is a minimal domain—eight positions—sufficient for the construction. The key pair is $(\textsc{MoBr}, \textsc{FaBr})$: mother's brother and father's brother.

3.2 Language L_D: Distinguished Encoding

Language L_D ("distinguished") assigns distinct expressions to distinct kinship positions.

Definition 14 (Distinguished Coding Scheme)**.** *L_D over $\mathcal{A}_{\text{kin}}$ has:*

- *Alphabet $\Sigma_D = \{\tau_{\textsc{Mo}}, \tau_{\textsc{Fa}}, \tau_{\textsc{MoBr}}, \tau_{\textsc{FaBr}}, \tau_{\textsc{MoSi}}, \tau_{\textsc{FaSi}}, \tau_{\textsc{Br}}, \tau_{\textsc{Si}}\}$*
- *Expression sets: $E_D(a) = \{\tau_a\}$ for each $a \in \mathcal{A}_{\text{kin}}$*
- *Distribution: $P_D(\tau_a \mid a) = 1$*

Each position has exactly one expression: its dedicated token. This models languages like Tamil that lexically distinguish mother's brother (*māmā*) from father's brother (*periappa*).

3.3 Language L_C: Collapsed Encoding

Language L_C ("collapsed") merges some kinship positions into shared expressions.

Definition 15 (Collapsed Coding Scheme)**.** *L_C over $\mathcal{A}_{\text{kin}}$ has:*

- *Alphabet $\Sigma_C = \{\tau_{\textsc{Mo}}, \tau_{\textsc{Fa}}, \tau_{\textsc{Uncle}}, \tau_{\textsc{Aunt}}, \tau_{\textsc{Br}}, \tau_{\textsc{Si}}\}$*
- *Expression sets:*

$$E_C(\textsc{MoBr}) = E_C(\textsc{FaBr}) = \{\tau_{\textsc{Uncle}}\}$$
$$E_C(\textsc{MoSi}) = E_C(\textsc{FaSi}) = \{\tau_{\textsc{Aunt}}\}$$

  and $E_C(a) = \{\tau_a\}$ for $a \in \{\textsc{Mo}, \textsc{Fa}, \textsc{Br}, \textsc{Si}\}$

- *Distribution: $P_C(x \mid a) = 1$ for $x \in E_C(a)$*

This models languages like English that collapse mother's brother and father's brother into "uncle."

3.4 Distributional Identity

Both languages generate corpora from the same distribution over kinship positions.

Definition 16 (Kinship Distribution)**.** *Let $P_{\mathcal{A}}$ be a distribution over $\mathcal{A}_{\text{kin}}$ reflecting natural usage frequencies. For concreteness:*

$$P_{\mathcal{A}}(\textsc{Mo}) = P_{\mathcal{A}}(\textsc{Fa}) = 0.2, \quad P_{\mathcal{A}}(\textsc{Br}) = P_{\mathcal{A}}(\textsc{Si}) = 0.15$$

$$P_{\mathcal{A}}(\textsc{MoBr}) = P_{\mathcal{A}}(\textsc{FaBr}) = P_{\mathcal{A}}(\textsc{MoSi}) = P_{\mathcal{A}}(\textsc{FaSi}) = 0.075$$

The exact values don't matter for the existence result. What matters is that both L_D and L_C generate corpora by sampling positions from this same distribution and encoding them according to their respective schemes.

3.5 The Witness Pair

The key observation is immediate from the definitions.

Proposition 17 (Canonical Expressions Differ)**.** *For the pair* $(\textsc{MoBr}, \textsc{FaBr})$*:*

$$x_D^*(\textsc{MoBr}) = \tau_{\textsc{MoBr}} \neq \tau_{\textsc{FaBr}} = x_D^*(\textsc{FaBr})$$
$$x_C^*(\textsc{MoBr}) = \tau_{\textsc{Uncle}} = x_C^*(\textsc{FaBr})$$

In L_D, the canonical expressions for mother's brother and father's brother are distinct tokens. In L_C, they are the same token. This is the structural difference that drives the geometric divergence.

3.6 Contextual Differentiation

For the proof to work, we need that the corpus in L_D provides evidence distinguishing MoBr from FaBr. This is where the anchored domain's structure matters.

In natural language use, mother's brother and father's brother appear in different contexts:

- "My MoBr came from my mother's side of the family."
- "My FaBr looks just like my father."
- Cross-culturally, MoBr and FaBr often have different social roles (maternal uncle as nurturer vs. paternal uncle as authority figure in many societies).

We formalize this as a condition on corpus structure.

Definition 18 (Contextual Differentiation)**.** *A corpus* $\mathcal{C}$ contextually differentiates *positions* $a_1, a_2 \in \mathcal{A}$ *if the distribution of contexts in which* a_1 *appears differs from the distribution of contexts in which* a_2 *appears.*

Assumption 19 (Differentiated Corpus)**.** *The corpus* $\mathcal{C}_D$ *generated by* L_D *contextually differentiates* $\textsc{MoBr}$ *from* $\textsc{FaBr}$*.*

This assumption is mild. It says that the underlying semantic difference between mother's brother and father's brother is reflected in usage patterns. If this assumption failed—if the two positions were used in statistically identical contexts—then they would be semantically indistinguishable by any behavioral criterion, and the question of whether a learner "distinguishes" them would be meaningless.

3.7 Richer Constructions

The collapsed/distinguished construction is minimal. Richer constructions are possible:

Compositional encoding. Instead of atomic tokens, L_D could encode positions compositionally:

$$x_D(\textsc{MoBr}) = \textsc{Maternal} \cdot \textsc{Uncle}$$
$$x_D(\textsc{FaBr}) = \textsc{Paternal} \cdot \textsc{Uncle}$$

This creates intermediate states (after processing MATERNAL but before UNCLE) that affect path structure. We return to this in Section 6 on geodesics.

Partial collapse. Languages can collapse some distinctions while preserving others. This creates complex geometric patterns where some pairs are equidistant across languages and others diverge.

Frequency variation. Even with identical lexical inventories, different usage frequencies can affect geometry. A language that discusses maternal kin more frequently will develop different representational structure than one emphasizing paternal kin.

For the existence proof, the simple collapsed/distinguished construction suffices. The richer constructions show that the phenomenon is not limited to one special case.

4. The Key Lemma

The heart of the proof is connecting coding structure to predictive behavior. We show that an accurate learner *must* develop different output distributions for tokens that appear in different contexts.

4.1 Predictive Differentiation

Lemma 20 (Competence Implies Differentiation)**.** *Let T be an ε-competent learner on corpus $\mathcal{C}$. Let τ_1, τ_2 be distinct tokens in $\mathcal{C}$. If the corpus contextually differentiates τ_1 from τ_2 with divergence $\delta > 0$, and $\varepsilon < \varepsilon_0(\delta)$ for a threshold function ε_0, then:*

$$d\big(h(\tau_1), h(\tau_2)\big) > 0$$

Proof sketch. Let D_1 be the true distribution of continuations following τ_1 in the corpus, and D_2 the distribution following τ_2. Contextual differentiation means $\mathrm{JS}(D_1, D_2) = \delta > 0$.

Suppose, for contradiction, that $h(\tau_1) = h(\tau_2)$. Then the learner uses the same output distribution Q after both tokens.

The cross-entropy on τ_1 contexts is $H(D_1, Q)$, and on τ_2 contexts is $H(D_2, Q)$. The optimal choice—using D_1 after τ_1 and D_2 after τ_2—achieves entropy $H(D_1)$ and $H(D_2)$ respectively.

The excess cross-entropy from using Q for both is:

$$\Delta = p_1 \cdot D_{\mathrm{KL}}(D_1 \| Q) + p_2 \cdot D_{\mathrm{KL}}(D_2 \| Q)$$

where p_1, p_2 are the frequencies of τ_1, τ_2 in the corpus.

The minimum excess over all choices of Q is achieved by the frequency-weighted mixture and is bounded below by a function of $\mathrm{JS}(D_1, D_2)$:

$$\Delta_{\min} \geq f(\delta, p_1, p_2) > 0$$

For $\varepsilon < \Delta_{\min}$, no learner using identical distributions for τ_1 and τ_2 can be ε-competent. Therefore, an ε-competent learner must have $p_T(\cdot \mid h(\tau_1)) \neq p_T(\cdot \mid h(\tau_2))$, which implies $d(h(\tau_1), h(\tau_2)) > 0$. $\square$

The lemma formalizes an intuition: if a learner is good at prediction, it must represent predictively-relevant distinctions. Tokens that predict different futures must occupy different representational states.

4.2 Application to the Construction

In language L_D, the tokens τ_{MoBr} and τ_{FaBr} are distinct and (by Assumption 19) contextually differentiated. By Lemma 20, any ε-competent learner satisfies:

$$d_D(\text{MoBr}, \text{FaBr}) = d\big(h(\tau_{\text{MoBr}}), h(\tau_{\text{FaBr}})\big) > 0$$

In language L_C, both positions map to the same token τ_{Uncle}:

$$h_C(\text{MoBr}) = h(\tau_{\text{Uncle}}) = h_C(\text{FaBr})$$

Same string, same internal state, same output distribution:

$$d_C(\text{MoBr}, \text{FaBr}) = d\big(h(\tau_{\text{Uncle}}), h(\tau_{\text{Uncle}})\big) = 0$$

4.3 The Collapsed-Token Triviality

The L_C case is almost trivially true: if two meanings share an expression, a learner processing that expression reaches a single state, and the distance is zero by definition.

This "triviality" is precisely the point. The existence result shows that *even this minimal structural difference*—collapsing two meanings into one token versus keeping them distinct—is sufficient to induce geometric divergence. No sophisticated argument about representations is needed. The effect follows directly from the coding structure.

5. The Non-Isometry Result

We now complete the proof of Theorem 1.

5.1 The Witness

Proposition 21 (Non-Isometry)**.** *Let T_D be an ε-competent learner on $\mathcal{C}_D$ and T_C be an ε-competent learner on $\mathcal{C}_C$, with ε sufficiently small. Then:*

$$d_D(\mathit{MoBr}, \mathit{FaBr}) > 0 = d_C(\mathit{MoBr}, \mathit{FaBr})$$

Proof. By Lemma 20 applied to L_D: since $\tau_{\text{MoBr}} \neq \tau_{\text{FaBr}}$ and the corpus contextually differentiates them, $d_D(\text{MoBr}, \text{FaBr}) > 0$.

In L_C: both meanings have canonical expression τ_{Uncle}. The learner's internal state after processing this token is $h(\tau_{\text{Uncle}})$ regardless of the underlying meaning. Therefore $h_C(\text{MoBr}) = h_C(\text{FaBr})$, and $d_C(\text{MoBr}, \text{FaBr}) = 0$. $\square$

5.2 What Non-Isometry Means

Two metric spaces (X, d_X) and (Y, d_Y) are *isometric* if there exists a bijection $\phi : X \to Y$ such that $d_Y(\phi(x_1), \phi(x_2)) = d_X(x_1, x_2)$ for all $x_1, x_2 \in X$. Isometric spaces have identical geometric structure—they differ only in labeling.

The induced geometries $(\mathcal{A}_{\text{kin}}, d_D)$ and $(\mathcal{A}_{\text{kin}}, d_C)$ are *not* isometric. Any candidate isometry ϕ must satisfy:

$$d_C(\phi(\text{MoBr}), \phi(\text{FaBr})) = d_D(\text{MoBr}, \text{FaBr}) > 0$$

But in $(\mathcal{A}_{\text{kin}}, d_C)$, the only pairs with distance zero are those sharing a lexeme: $\{(\text{MoBr}, \text{FaBr}), (\text{MoSi}, \text{FaSi})\}$ and self-pairs. The only pairs with positive distance are those with distinct lexemes. There is no pair in $(\mathcal{A}_{\text{kin}}, d_C)$ that can serve as the image of $(\text{MoBr}, \text{FaBr})$ under a distance-preserving map.

The geometries are structurally different. They cannot be aligned by any relabeling.

5.3 Interpretation

The same eight kinship positions, anchored by the same genealogical structure, induce *different patterns of representational distance* depending on the coding scheme.

In L_D, mother's brother and father's brother are representationally distant—they predict different linguistic futures.

In L_C, they are representationally identical—collapsed into a single predictive state.

An agent trained on L_D and an agent trained on L_C, presented with the "same" kinship scenario (same family tree, same individuals), will have different internal geometries. Questions like "how similar are these two relatives?" have different answers depending on the agent's linguistic training—not because the agents have different knowledge, but because they have different representational structure.

This is the precise sense in which coding scheme shapes cognitive geometry.

5.4 The Triviality Objection

A hostile reader might object: the collapsed construction is definitionally trivial. Of course if two meanings share the same canonical string, the learner reaches the same internal state, and the distance is zero. This is not a deep result about representation—it's an artifact of how we defined $h_L(a)$.

We offer two responses.

First: for a pure existence proof, triviality is a feature. The theorem claims that coding-induced geometric divergence is *possible*. The collapsed construction exhibits this with minimal machinery. Even the simplest structural difference—lexical collapse—suffices. The proof is complete.

Second: the collapsed construction is a limiting case. A stronger, non-degenerate result is available. We develop it in the next section.

6. The Strong Result: Compositional vs. Lexicalized Geometry

The collapsed/distinguished construction yields $d > 0$ versus $d = 0$. This is sufficient for existence but feels definitionally thin. We now present a stronger result: two languages that *both* distinguish all meanings, yet induce geometries with different *structure*.

6.1 The Compositional Construction

Definition 22 (Lexicalized Coding Scheme L_{lex})**.** *L_{lex} over $\mathcal{A}_{\text{kin}}$ assigns an atomic token to each position:*

$$\Sigma_{\text{lex}} = \{\tau_{Mo}, \tau_{Fa}, \tau_{MoBr}, \tau_{FaBr}, \tau_{MoSi}, \tau_{FaSi}, \tau_{Br}, \tau_{Si}\}$$

with $E_{\text{lex}}(a) = \{\tau_a\}$ and $P_{\text{lex}}(\tau_a \mid a) = 1$.

This is identical to L_D from the minimal construction. Each meaning has a dedicated, atomic token.

Definition 23 (Compositional Coding Scheme L_{comp})**.** *L_{comp} over $\mathcal{A}_{\text{kin}}$ uses feature morphemes:*

$$\Sigma_{\text{comp}} = \{\textsc{Mat}, \textsc{Pat}, \textsc{Par}, \textsc{Sib}, M, F\}$$

(Maternal, Paternal, Parent, Sibling, Male, Female). Expressions are concatenations:

$$\begin{aligned}
E_{\text{comp}}(\textsc{Mo}) &= \{\textsc{Par} \cdot F\} \\
E_{\text{comp}}(\textsc{Fa}) &= \{\textsc{Par} \cdot M\} \\
E_{\text{comp}}(\textsc{MoBr}) &= \{\textsc{Mat} \cdot \textsc{Par} \cdot \textsc{Sib} \cdot M\} \\
E_{\text{comp}}(\textsc{FaBr}) &= \{\textsc{Pat} \cdot \textsc{Par} \cdot \textsc{Sib} \cdot M\} \\
E_{\text{comp}}(\textsc{MoSi}) &= \{\textsc{Mat} \cdot \textsc{Par} \cdot \textsc{Sib} \cdot F\} \\
E_{\text{comp}}(\textsc{FaSi}) &= \{\textsc{Pat} \cdot \textsc{Par} \cdot \textsc{Sib} \cdot F\} \\
E_{\text{comp}}(\textsc{Br}) &= \{\textsc{Sib} \cdot M\} \\
E_{\text{comp}}(\textsc{Si}) &= \{\textsc{Sib} \cdot F\}
\end{aligned}$$

Both languages distinguish all eight positions. Neither collapses any meanings. The difference is purely structural: L_{lex} uses atomic symbols; L_{comp} uses compositional expressions built from shared morphemes.

6.2 Intermediate States and Hub Structure

The key observation: processing a compositional expression creates intermediate internal states.

In L_{lex}, processing τ_{MoBr} yields a single state $h(\tau_{\text{MoBr}})$. No intermediates.

In L_{comp}, processing $\textsc{Mat} \cdot \textsc{Par} \cdot \textsc{Sib} \cdot \text{M}$ yields a sequence of states:

$$h_0 \to h(\textsc{Mat}) \to h(\textsc{Mat} \cdot \textsc{Par}) \to h(\textsc{Mat} \cdot \textsc{Par} \cdot \textsc{Sib}) \to h(\textsc{Mat} \cdot \textsc{Par} \cdot \textsc{Sib} \cdot \text{M})$$

These intermediate states are *shared* across meanings:

- $h(\textsc{Mat})$ lies on the path to both $\textsc{MoBr}$ and $\textsc{MoSi}$
- $h(\textsc{Par} \cdot \textsc{Sib})$ lies on the path to all uncle/aunt terms
- $h(\textsc{Sib} \cdot \text{M})$ lies on the path to both $\textsc{Br}$ and (as a suffix) the uncle terms

Definition 24 (Hub State)**.** *A state $h^* \in \mathcal{H}$ is a* hub *for meaning set $S \subseteq \mathcal{A}$ if h^* lies on the processing path for every $a \in S$: for all $a \in S$, there exists a prefix p of $x^*(a)$ such that $h(p) = h^*$.*

Proposition 25 (Hub Existence)**.** *In L_{comp}, the state $h(\textsc{Par}{\cdot}\textsc{Sib})$ is a hub for $\{\textsc{MoBr}, \textsc{FaBr}, \textsc{MoSi}, \textsc{FaSi}\}$.*

In L_{lex}, no non-trivial hub exists for any subset of size ≥ 2.

Proof. In L_{comp}: all four uncle/aunt expressions contain $\textsc{Par}\cdot\textsc{Sib}$ as a substring. The learner passes through $h(\textsc{Par} \cdot \textsc{Sib})$ when processing any of them.

In L_{lex}: each meaning has a single-token expression. The only shared prefix is the empty string, yielding the initial state h_0, which is trivial. □

6.3 Processing Paths vs. Geodesics

We must distinguish two notions of "path" through representational space.

Definition 26 (Processing Path)**.** *The* processing path *to meaning a in language L is the sequence of internal states traversed while processing the canonical expression $x^*(a) = \sigma_1 \cdot \sigma_2 \cdots \sigma_n$:*

$$\pi_L(a) = \big(h_0, h(\sigma_1), h(\sigma_1 \cdot \sigma_2), \ldots, h(\sigma_1 \cdots \sigma_n)\big)$$

This is the trajectory through $\mathcal{H}$ during incremental comprehension.

Definition 27 (Geodesic)**.** *A* geodesic *from h_1 to h_2 in $(\mathcal{H}, d)$ is a path $\gamma : [0,1] \to \mathcal{H}$ with $\gamma(0) = h_1$, $\gamma(1) = h_2$, and minimal length under the behavioral metric.*

These are different objects. The processing path is determined by the linguistic structure of the expression—it's the route you *actually take* when parsing. The geodesic is the shortest route in the abstract metric space—it may or may not correspond to any linguistic expression.

The hub $h(\textsc{Par} \cdot \textsc{Sib})$ lies on the *processing path* to all uncle/aunt meanings. This is true by construction. But does it lie on the *geodesic* between, say, $h(\textsc{MoBr})$ and $h(\textsc{FaBr})$? That's a separate question, and the answer depends on the geometry of the learned representation space in ways we cannot determine a priori.

Proposition 28 (Processing Paths Differ)**.** *In L_{comp}: the processing paths to $\textsc{MoBr}$, $\textsc{FaBr}$, $\textsc{MoSi}$, $\textsc{FaSi}$ all pass through the hub state $h(\textsc{Par} \cdot \textsc{Sib})$.*

In L_{lex}: the processing paths to these meanings are single-step jumps from h_0 to the final state. No shared intermediate is visited.

Proof. Immediate from the expression structures. In L_{comp}, all four expressions contain the substring $\textsc{Par} \cdot \textsc{Sib}$. In L_{lex}, each expression is a single token. □

This is a structural difference in *how meanings are accessed*, not (directly) in the static distances between them. The compositional language forces comprehension through shared intermediate states; the lexicalized language does not.

6.4 When Processing Paths Approximate Geodesics

Under what conditions do processing paths correspond to geodesics?

If the behavioral metric varies smoothly along processing paths—if each token-step produces a small, incremental change in output distribution—then the processing path is approximately a geodesic. The route you take during parsing is close to the shortest route in the metric space.

This is plausible for well-trained learners on natural-like corpora. Each morpheme contributes incrementally to the predicted meaning; the output distribution evolves gradually. Under such conditions, the hub states that lie on processing paths would also lie near geodesics.

But we state this as a modeling assumption, not a theorem:

Assumption 29 (Smooth Processing)**.** *The behavioral metric varies smoothly along processing paths: for consecutive states h_i, h_{i+1} on a processing path, $d(h_i, h_{i+1}) \leq \delta$ for some small δ.*

Under Assumption 29, the compositional hub structure becomes geometrically relevant: hubs lie not just on processing paths but near geodesics. Without this assumption, we can only claim the weaker result about processing dynamics.

6.5 The Strong Non-Isometry

Theorem 30 (Structural Non-Isometry)**.** *The representational structures induced by L_{lex} and L_{comp} differ:*

1. ***No collapse****: Both have $d(a_1, a_2) > 0$ for all distinct $a_1, a_2 \in \mathcal{A}_{\text{kin}}$.*
2. ***Hub structure****: The reachable state space $\mathcal{H}_{\text{comp}|}$ contains non-trivial hub states; $\mathcal{H}_{\text{lex}}$ does not.*
3. ***Processing dynamics****: Accessing uncle/aunt meanings in L_{comp} traverses shared intermediate states; in L_{lex} it does not.*
4. ***Geodesics*** *(conditional): Under Assumption 29, geodesics in $\mathcal{H}_{\text{comp}}$ route through hubs; geodesics in $\mathcal{H}_{\text{lex}}$ are direct jumps.*

Claims (1)–(3) hold unconditionally. Claim (4) requires the smooth processing assumption.

This is stronger than the collapsed construction. Both languages distinguish all meanings—no "trivial" $d = 0$. Yet the structures differ: different hub states, different processing dynamics, and (under smoothness) different geodesic routing.

6.6 Shared Structure and Distance

Hub existence also affects pairwise distances, though the argument is subtler.

In L_{comp}, expressions for related meanings share morphemes. The expressions for MoBr and MoSi both begin with Mat · Par · Sib; they differ only in the final gender morpheme. An ε-competent learner, after processing the shared prefix, is in a state that could continue to either meaning. The output distributions for the complete expressions share this common ancestry.

In L_{lex}, the tokens τ_{MoBr} and τ_{MoSi} are unrelated atoms. Their output distributions are learned independently, shaped only by their respective contexts in the corpus.

Conjecture 31 (Shared Structure Reduces Distance)**.** *For meanings a_1, a_2 that share a compositional prefix in L_{comp}:*

$$d_{\text{comp}}(a_1, a_2) < d_{\text{lex}}(a_1, a_2)$$

Compositional encoding, by forcing shared intermediate states, makes related meanings representationally closer.

We state this as a conjecture rather than a theorem because the inequality depends on corpus statistics and learner behavior in ways that require more detailed analysis. The structural claims (hub existence, processing path differences) are proven; the metric claims are plausible but not established.

6.7 Implications for Cognition

The structural differences have cognitive implications, primarily through *processing dynamics* rather than static distances.

When a speaker of L_{comp} comprehends "mother's brother," they traverse the hub state $h(\text{PAR}\cdot\text{SIB})$—a state that is also visited when processing "father's brother," "mother's sister," and "father's sister." The shared structure is *activated during comprehension*. This creates opportunities for priming, interference, and analogical connection that arise naturally from the processing path.

When a speaker of L_{lex} comprehends the atomic token for "mother's brother," they jump directly to the final state. No shared intermediate is visited. The connection to "father's brother" (both are uncles) is not *traversed*—it must be *inferred* separately.

This suggests compositional and lexicalized languages support different *styles* of thought:

- **Compositional**: comprehension naturally decomposes into features; shared structure is activated during processing; analogies along feature dimensions are cheap because the relevant intermediates are visited.
- **Lexicalized**: concepts are accessed holistically; connections between meanings require explicit bridging; the processing space is more fragmented.

Neither style is superior. They are different cognitive affordances, shaped by the coding scheme. The compositional style makes certain connections automatic; the lexicalized style makes certain distinctions sharper. The geometry of thought—in the sense of what is easy versus hard to think—differs.

6.8 Operationalization

The theoretical constructs admit empirical measurement.

Behavioral distance. The metric $d(h_1, h_2) = \text{JS}(p(\cdot|h_1), p(\cdot|h_2))$ is directly observable: present anchor expressions to a trained model, read off the output distributions (logits), compute divergence. No access to internal representations required.

Hub detection. To test whether a state h^* functions as a hub, probe the model's internal activations (e.g., residual stream in a transformer) at the position corresponding to the candidate hub prefix. If activations cluster similarly across expressions that share the prefix but diverge afterward, the hub structure is present. This requires interpretability tools beyond output-layer measurement, but the target is well-defined.

Distance comparisons. The conjecture that $d_{\text{comp}}(a_1, a_2) < d_{\text{lex}}(a_1, a_2)$ for prefix-sharing meanings is testable: train models on compositional vs. lexicalized corpora, measure pairwise distances on anchored meanings, compare. The prediction is directional and quantitative.

Smooth processing. Assumption 29 can be checked empirically: measure $d(h_i, h_{i+1})$ along processing paths. If consecutive states have small divergence, the assumption holds; if there are discontinuous jumps, it fails.

A companion paper develops the full experimental methodology, including protocols for cross-linguistic comparison using models trained on natural language corpora. The present paper establishes that the theoretical constructs are coherent and the predictions are well-posed; the empirical program tests whether natural languages instantiate the sufficient conditions.

7. Scope and Implications

We now clarify what the theorem establishes and what it does not.

7.1 What Is Established

Existence. There exist coding schemes over a shared anchored domain such that any sufficiently accurate predictive learner induces non-isometric representational geometries. Strong linguistic relativity—in the precise sense of coding-induced geometric divergence—is mathematically coherent and constructively realizable.

Identifiability. The geometric divergence is observable under behavioral metrics. We do not need access to internal representations or coordinate systems. The distance function is defined by predictive behavior: how differently two states forecast continuations. This makes the non-isometry measurable, not merely posited.

Architecture independence. The result holds for *any* ε-competent learner. Transformers, RNNs, n-gram models—the architecture does not matter. What matters is predictive accuracy. If a system models the corpus distribution well, it inherits the geometric structure forced by the coding scheme.

Sufficient conditions. The construction identifies conditions sufficient for geometric divergence:

1. An anchored domain with externally-defined identity criteria
2. Two coding schemes covering the domain
3. Lexical collapse in one scheme (multiple meanings share an expression)
4. Lexical distinction in the other (different meanings have different expressions)
5. Contextual differentiation (distinguished meanings appear in different contexts)

When these conditions hold, non-isometry follows. The proof is constructive: we exhibit the witness.

7.2 What Is Not Claimed

Not an empirical claim about natural languages. We do not assert that English and Tamil speakers inhabit non-isometric cognitive geometries. We construct toy languages and prove geometric divergence for those languages. Whether natural languages satisfy the construction's conditions is an empirical question we do not answer here.

Not cognitive equivalence. The theorem concerns predictive learners trained on text. We do not claim that such learners are minds, or that their representational geometry equals human cognitive geometry. The relevance to human cognition depends on a bridge assumption: that model geometry proxies cognitive geometry to some approximation. This assumption is testable but not proven.

Not universal. Not all language differences produce geometric differences. Grammar, phonology, word order—these may leave geometry invariant. The theorem identifies *lexicalization patterns* as the relevant variable. Two languages with identical lexicalization but different syntax might induce isometric geometries. The construction is specific about what matters.

Not determinism. Speakers are not prisoners of their language's geometry. Humans use scaffolding—paraphrase, metaphor, borrowing, formal notation—to transcend the default paths their language provides. The geometry defines what is *cheap*, not what is *possible*. A speaker of L_C can distinguish mother's brother from father's brother; they simply lack a lexical shortcut for doing so.

7.3 The Reframed Question

The Sapir-Whorf hypothesis has been debated for a century. Does language shape thought? The question is too vague to answer. "Shape" and "thought" admit too many interpretations.

The existence result reframes the question precisely:

> *Do natural languages differ in ways that satisfy the conditions of the construction?*

This is answerable. The conditions are:

- **Anchored domain**: a semantic space with language-independent structure (kinship, color, space, evidentiality)
- **Lexicalization difference**: one language collapses distinctions the other preserves
- **Contextual differentiation**: the distinguished meanings have different usage distributions

Linguistic typology documents these differences. Tamil distinguishes maternal from paternal uncle; English collapses them. Russian distinguishes light from dark blue; English collapses them. Hopi grammaticalizes evidentiality; English leaves it optional.

The empirical question is not "do these differences exist?"—they manifestly do. The question is whether the differences satisfy the conditions at scale, in naturally-occurring corpora, to a degree sufficient to produce measurable geometric divergence in trained models.

This is testable. Train models on monolingual corpora. Measure geometric properties. Compare across languages. If lexicalization patterns predict geometric patterns, the construction applies to natural languages. If not, the construction identifies a theoretical possibility that natural languages don't instantiate.

7.4 Implications

For linguistics. Typological features become geometric predictors. The question "how does Tamil kinship terminology differ from English?" becomes "what geometric properties does Tamil kinship space have that English kinship space lacks?"

For cognitive science. The strong/weak Whorf distinction loses its force. Relativity is not binary but continuous: how much geometric divergence, on which domains, between which languages?

These are quantitative questions with empirical answers.

For AI. Training data shapes representational geometry. A model trained on English develops different geometric structure than one trained on Tamil. Multilingual training expands the geometry by adding paths and reducing barriers between concepts. Monolingual models may have systematic blind spots—regions of meaning space that are harder to reach because the training language lacks short paths to them.

7.5 From Existence to Measurement

The existence result establishes possibility and identifies sufficient conditions. What remains is to test whether natural languages satisfy those conditions.

The empirical program:

1. Select anchored domains with known cross-linguistic lexicalization differences
2. Train predictive models on monolingual corpora
3. Measure behavioral geometry (pairwise JS divergence on anchor expressions)
4. Test whether lexicalization patterns predict geometric patterns

If the predictions hold, we will have established that linguistic relativity occurs in specific domains to measurable degrees.

Coda

The Sapir-Whorf debate has persisted because the central question—does language shape thought?—was never operationalized precisely enough to admit a clear answer.

This paper contributes one precise version of the question and answers it. There exist coding schemes such that any competent predictive learner induces non-isometric representational geometries. Under behavioral metrics, this is a theorem.

The open question is empirical: do natural languages satisfy the construction's conditions? Lexicalization differences exist; the question is whether they produce measurable geometric divergence in trained models. A companion paper pursues this measurement program.